

How to calculate Correlation Coefficient

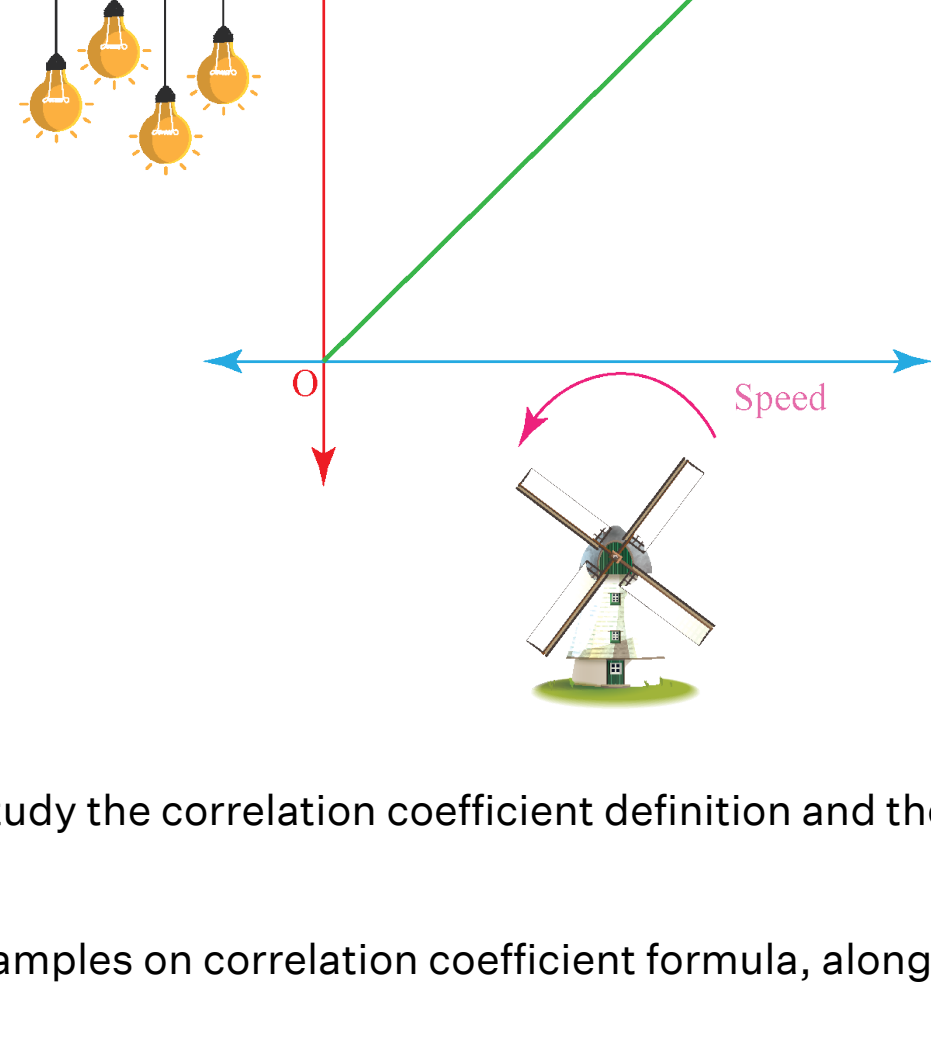
The word "correlation" is made by clubbing the words "co" and "relation". The word "co" means together, thus, correlation means the relationship between any set of data when considered together.

In Statistics, the correlation coefficient is a measure defined between the numbers -1 and +1 and represents the linear interdependence of the set of data.

Let us analyze the following situation:

For electricity generation using a windmill, if the speed of the wind turbine increases, the generation output will increase accordingly. Thus, the variable speed and electricity output have a positive correlation here.

This correlation can be studied using the correlation coefficient.



In this mini-lesson, we will study the correlation coefficient definition and the correlation coefficient formula.

Check out the interactive examples on correlation coefficient formula, along with practice questions at the end of the page.

Lesson Plan

1. What Is Correlation Coefficient?
2. Important Notes
3. Challenging Questions
4. Solved Examples
5. Interactive Questions

What Is Correlation Coefficient?

Correlation Coefficient Definition

The correlation coefficient is used to measure the strength of the relationship between two variables.

The value of the correlation coefficient ranges from -1.0 to +1.0

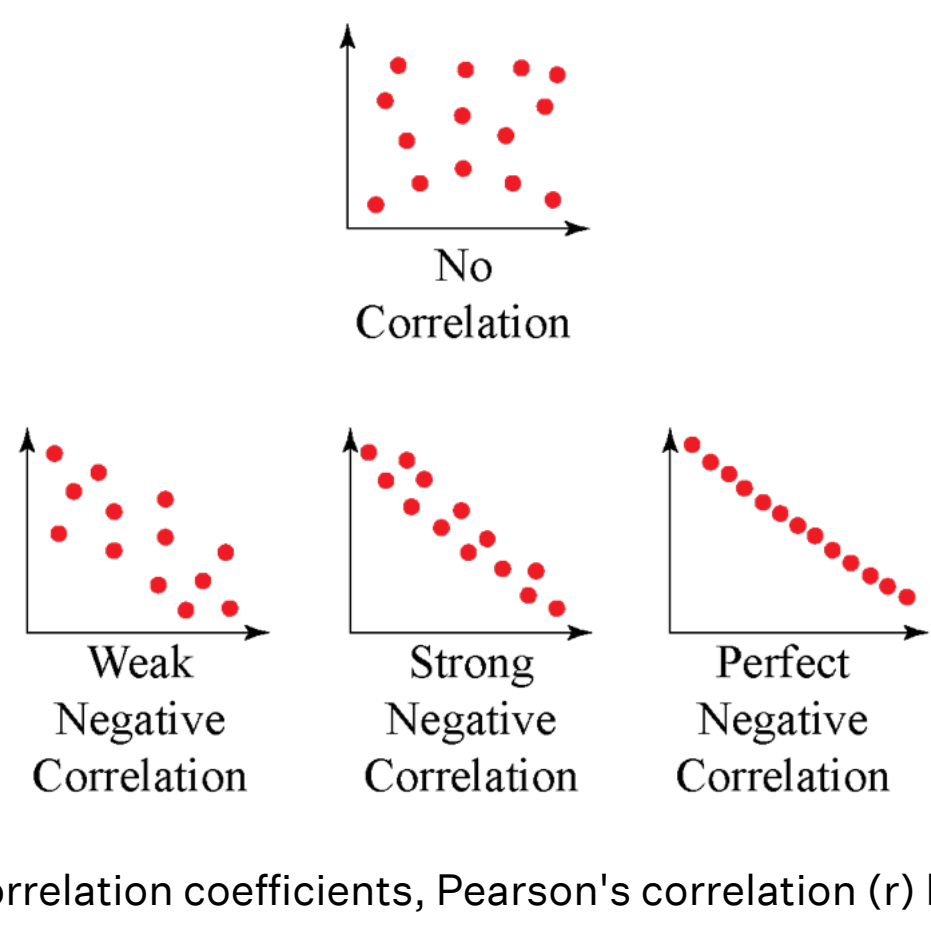
This means that any value beyond this range will be the result of an error in correlation measurement.

A correlation of value -1.0 means a perfect negative correlation, while a correlation of +1.0 means a perfect positive correlation.

A correlation of 0.0 means no linear relationship between the movement of the two variables.

Correlation Coefficient Example

The following correlation graphs show the examples of different range of values for a correlation coefficient:



There are several types of correlation coefficients, Pearson's correlation (r) being the most common among all.

It measures the strength and direction of the linear relationship between the two variables and cannot capture nonlinear relationships between two variables.

It cannot differentiate between dependent and independent variables.

How to Calculate Correlation Coefficient?

- The correlation coefficient can be calculated by first determining the covariance of the given variables.
- This value is then divided by the product of standard deviations for these variables.
- The equation given below summarizes the above concept:

$$\rho_{xy} = \frac{\text{Cov}(x,y)}{\sigma_x \sigma_y}$$

where,

ρ_{xy} = Pearson's product-moment correlation coefficient

$\text{Cov}(x,y)$ = covariance of variables x and y

σ_x = standard deviation of x

σ_y = standard deviation of y

Covariance

Covariance gives the joint relationship between two random variables.

It is calculated using the following formula:

$$\text{Cov}(X, Y) = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{n}$$

where,

X, Y = random variables

X_i = data value of x

Y_i = data value of y

$\bar{X}$ = mean of all values of X

$\bar{Y}$ = mean of all values of Y

n = Total number of values of X or Y

Formula of Correlation Coefficient

The formula for correlation coefficient is given as:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

where,

n = Quantity of information

$\sum x$ = Total of all values for first variable

$\sum y$ = Total of all values for second variable

$\sum xy$ = Sum of product of first and second value

$\sum x^2$ = Sum of squares of the first value

$\sum y^2$ = Sum of squares of the second value

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$



Important Notes

- The value of the correlation coefficient ranges from -1.0 to +1.0.
- Correlation is symmetric for both variables, say x, y .
- For a positive correlation: the values increase together.
- For a negative correlation: one value decreases as the other increases.

Types of Correlation Coefficient Formulas

There are different correlation formulas available for the calculation of coefficient formulas:

Pearson's Correlation Coefficient Formula

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

where,

n = Quantity of information

$\sum x$ = Total of all values for first variable

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$\sum x^2$ = Sum of squares of the first value

$\sum y^2$ = Sum of squares of the second value

Linear Correlation Coefficient Formula

$$r_{xy} = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{\sqrt{[n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2][n \sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2]}}$$

where the symbols for variables have their general meaning.

Sample Correlation Coefficient Formula

$$r_{xy} = \frac{S_{xy}}{S_x S_y}$$

where,

S_x, S_y = Standard deviations

S_{xy} = Sample covariance

Population Correlation Coefficient Formula

$$\rho_{xy} = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

where,

σ_x, σ_y = Population standard deviations

σ_{xy} = Population covariance

What is the difference between Correlation and Regression?

Correlation	Regression
This indicates the nature and the extent of the linear relationship.	This studies the impact of the independent variable on the dependent variable.
The variables are positively or negatively correlated if the correlation is a positive or negative value respectively.	For a positive regression coefficient: For every unit increase in x , there is a corresponding average increase in y in b_{yx} . For a negative regression coefficient: For every unit increase in x , there is a corresponding average decrease in y in b_{yx} .
It is symmetric for both variables, say x, y .	It is not symmetric for both variables, say for variables x and y , b_{xy} and b_{yx} have different interpretations and meaning.



Challenging Question

- How can you prove the symmetry of correlation and non-symmetry for covariance?

Solved Examples

Example 1

Can you help Jake calculate the correlation coefficient for the following data?

$X = 4, 8, 12, 16$ and

$Y = 7, 14, 21, 28$

Solution

Given variables are:

$X = 4, 8, 12, 16$

$Y = 7, 14, 21, 28$

To find the linear coefficient of this data, we will first construct a table to get the required values of the formula:

X	Y	X^2	Y^2	XY
4	7	16	49	28
8	14	64	196	112
12	21	144	441	252
16	28	256	784	448
$\sigma X = 40$	$\sigma Y = 70$	$\sigma X^2 = 480$	$\sigma Y^2 = 1,470$	$\sigma XY = 840$

We know,

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Putting all values, we get:

$$\begin{aligned} r &= \frac{4 \times 840 - (40)(70)}{\sqrt{[4 \times 480 - (40)^2][4 \times 1,470 - (70)^2]}} \\ &= \frac{3,360 - 2,800}{\sqrt{[1,920 - 1,600][5,880 - 4,900]}} \\ &= \frac{560}{560} \\ &= 1 \end{aligned}$$

Hence, the Correlation coefficient:

$$r = 1$$

∴ Correlation coefficient, $r = 1$

Example 2

What is the difference between covariance and correlation for any given variables?

Solution

The difference between covariance and correlation can be studied using the following table:

Correlation	Covariance
Correlation is the measure to indicate the strength of the relationship between two variables.	Covariance is the measure to indicate the extent up to which two variables can change.
It is the scaled form of correlation.	It is a measure of correlation.
It lies between -1 to +1.	It lies between $-\infty$ to $+\infty$.
Change in scale does not affect correlation.	Change in scale affects the value of covariance.

Let's Summarize

The mini-lesson targeted the fascinating concept of the correlation coefficient. The math journey around correlation coefficient started with what a student already knew and went on to creatively crafting a fresh concept in the young minds. Done in a way that not only it is relatable and easy to grasp, but also will stay with them forever.

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FAQs on Correlation Coefficient

1.What is Pearson's Correlation?

Pearson's correlation is the measure of strength between any two variables. It is determined using the Pearson's correlation coefficient, whose values lie between -1 and +1

The formula to calculate Pearson's correlation coefficient is given by:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

where,

n = Quantity of information

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2.What are the potential problems to Pearson's Correlation?

Pearson's correlation can be used to measure the strength between any two variables. It is determined using the Pearson's correlation coefficient, whose values lie between -1 and +1

3.How to find Pearson's Correlation Coefficients?

The formula to calculate Pearson's correlation coefficient is given by:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

where,

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$\sum x$ = Total of all values for first variable

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$\sum xy$ = Sum of product of first and second value

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4.What is the Linear Correlation Coefficient?

Linear Correlation Coefficient is the measure of strength between any two variables. It is determined using the Pearson's correlation coefficient, whose values lie between -1 and +1

The formula to calculate Linear Correlation Coefficient is given by:

$$r_{xy} = \frac{n \sum_{i=1}^n x_i y_i - \sum_{i=1}^n x_i \sum_{i=1}^n y_i}{\sqrt{[n \sum_{i=1}^n x_i^2 - (\sum_{i=1}^n x_i)^2][n \sum_{i=1}^n y_i^2 - (\sum_{i=1}^n y_i)^2]}}$$